End-to-End Food Delivery Orders & ETA Logistics Analysis

About Dataset

Context

The urban food delivery ecosystem is a complex network involving customers, restaurants, delivery partners, order values, preparation times, traffic, weather, and delivery operations.

This dataset contains **50,000 food delivery orders** and is designed for analyzing delivery performance, operational efficiency, customer behavior, restaurant performance, and ETA prediction.

The dataset can be used for **Exploratory Data Analysis (EDA), SQL analysis, business intelligence dashboards, logistics analytics, and machine learning models for delivery time and delay prediction**.

Project Objective

The primary objective of this project is to develop an end-to-end food delivery analytics solution capable of:

- Analyzing customer ordering behavior and loyalty.
- Evaluating restaurant performance and preparation efficiency.
- Analyzing delivery driver experience and performance.
- Studying order values, discounts, taxes, and delivery fees.
- Identifying factors affecting delivery time.
- Analyzing traffic and weather impact on delivery operations.
- Measuring delivery delays and operational performance.
- Identifying high-risk delivery conditions.
- Detecting unusual delivery times and order values using statistical techniques.
- Building interactive dashboards for business decision-making.
- Preparing the dataset for ETA and delivery-delay prediction.

Dataset Structure & Files

The dataset contains **50,000 food delivery orders** covering customer information, restaurant details, driver operations, environmental conditions, financial metrics, and delivery performance.

Primary Business Domains

The dataset can be divided into four major domains:

1. **Customer Domain**
2. **Restaurant Domain**
3. **Driver & Delivery Domain**
4. **Environmental & Financial Domain**

What's Inside?

1. Customer Demographics & Loyalty

The dataset contains information about customers and their relationship with the food delivery service.

Possible information includes:

- Customer age
- Customer loyalty/tier
- Customer ordering behavior
- Customer segmentation
- Customer order frequency

This information can be used to understand different customer groups and their ordering patterns.

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2. Restaurant Logistics

The dataset contains restaurant-level information related to food preparation and restaurant performance.

Possible information includes:

- Restaurant
- Cuisine category
- Restaurant rating
- Historical preparation time

- Food preparation performance

This information helps analyze how restaurant operations influence overall delivery time.

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3. Driver Operations

The dataset contains information about delivery partners responsible for completing customer deliveries.

Possible information includes:

- Driver experience
- Driver performance rating
- Delivery partner characteristics
- Delivery performance

This can be used to understand whether driver experience and ratings are associated with delivery performance.

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4. Environmental & Financial Metrics

The dataset also contains operational and financial variables that influence food delivery.

Possible environmental factors include:

- Weather conditions
- Traffic density
- Delivery conditions

Financial information includes:

- Order subtotal
- Taxes
- Discounts
- Delivery fees
- Total order cost

These variables can be analyzed to understand the relationship between operating conditions, order economics, and delivery performance.

Project Title

End-to-End Food Delivery Orders & ETA Logistics Analysis

Project Overview

This project performs an end-to-end analysis of **50,000 food delivery orders**.

The project covers the complete data analytics lifecycle:

Raw Dataset
↓
Data Loading
↓
Data Understanding
↓
Data Cleaning
↓
Data Validation
↓
Exploratory Data Analysis
↓
Feature Engineering
↓
SQL Analysis
↓
Power BI Dashboard
↓
Business Insights
↓
ETA / Delay Prediction Preparation

The project combines **Python, SQL, and Power BI** to analyze customer behavior, restaurant operations, driver performance, delivery conditions, order economics, and ETA-related factors.

Project Type

End-to-End Data Analytics Project

Business Domain

Food Delivery / Logistics / E-Commerce / Operations Analytics

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Business Problem

Food delivery companies need to deliver orders quickly while maintaining good customer experience and controlling operational costs.

Delivery time can be affected by multiple factors such as:

- Restaurant preparation time
- Driver experience
- Driver performance
- Traffic
- Weather
- Customer behavior
- Order size
- Order value
- Delivery charges

Without proper analysis, it can be difficult to identify the major causes of delivery delays and operational inefficiencies.

Therefore, this project analyzes the complete food delivery order lifecycle to identify patterns in **orders, customers, restaurants, drivers, environmental conditions, delivery performance, and financial metrics**.

Project Objectives

The major objectives are:

1. Analyze overall food delivery order performance.
2. Understand customer demographics and loyalty behavior.
3. Analyze restaurant preparation performance.
4. Evaluate driver experience and ratings.
5. Analyze order value and financial components.
6. Study the effect of traffic on delivery time.
7. Study the effect of weather on delivery performance.
8. Identify delayed deliveries.
9. Analyze factors associated with ETA.

10. Detect statistical outliers.
11. Build SQL-based business analysis.
12. Create an interactive Power BI dashboard.
13. Prepare meaningful features for future ETA prediction.

Dataset Description

The dataset contains **50,000 records**, where each record represents a food delivery order.

The dataset combines information from different parts of the food delivery process.

Main Categories

Category	Purpose
Customer	Analyze customer demographics and loyalty
Restaurant	Analyze cuisine, ratings and preparation
Driver	Analyze driver experience and ratings
Environment	Analyze weather and traffic
Finance	Analyze subtotal, taxes, discounts and fees
Delivery	Analyze ETA and delivery performance

Dataset Features

The exact column names should be verified from the actual dataset file. The following represents the logical structure of the dataset.

Customer Features

- Customer information
- Customer age
- Customer loyalty tier

Restaurant Features

- Restaurant information
- Cuisine category

- Restaurant rating
- Historical preparation time

Driver Features

- Driver information
- Driver experience
- Driver rating

Environmental Features

- Weather condition
- Traffic density

Financial Features

- Order subtotal
- Tax amount
- Discount
- Delivery fee
- Total order amount

Delivery Features

- Estimated delivery time / ETA
- Actual delivery time, if available
- Delay indicator or delay duration, if available

> **Note:** Use the actual column names from the CSV file when implementing the project.

Data Dictionary

Customer Information

Feature	Description
Customer ID	Unique customer identifier
Age	Age of the customer
Loyalty Tier	Customer loyalty category

Restaurant Information

Feature	Description
Restaurant ID	Unique restaurant identifier
Cuisine	Type/category of food
Rating	Restaurant rating
Preparation Time	Historical food preparation duration

Driver Information

Feature	Description
Driver ID	Unique delivery partner identifier
Experience	Driver experience level
Rating	Driver performance rating

Environmental Information

Feature	Description
Weather	Weather condition during delivery
Traffic	Traffic density during delivery

Financial Information

Feature	Description
Subtotal	Value of food items before additional charges
Tax	Tax charged on the order
Discount	Discount applied
Delivery Fee	Delivery charge
Total Amount	Final customer order amount

Delivery Information

Feature	Description

ETA	Estimated delivery time
Delivery Time	Actual delivery duration, if available
Delay	Difference between expected and actual delivery performance

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Important Data Understanding Points

Before performing analysis, the following points should be checked:

1. Dataset Size

The dataset is expected to contain approximately:

50,000 orders

2. One Row Per Order

If the dataset contains one row per order, aggregation can be performed directly at order level.

3. Categorical Variables

Variables such as:

- Cuisine
- Loyalty Tier
- Weather
- Traffic

should be treated as categorical variables.

4. Numerical Variables

Variables such as:

- Age
- Rating
- Preparation Time
- Subtotal
- Discount
- Delivery Fee
- ETA

should be analyzed as numerical variables where applicable.

5. Target Variable

For ETA prediction, the target variable should be identified from the actual dataset.

For example:

ETA

or

Delivery Time

or

Delay

depending on the available columns.

Business Questions

General Analysis

1. How many orders are present?
2. How many unique customers are present?
3. How many restaurants are present?
4. How many drivers are present?
5. What is the overall order value?
6. What is the average order value?
7. What is the average delivery time?
8. What is the overall delivery performance?

Customer Analysis

9. Which loyalty tier generates the most orders?
10. Which customer group has the highest average order value?
11. What is the age distribution of customers?

12. Which age groups order most frequently?
13. Does loyalty level affect order value?
14. Which customer segments contribute the most revenue?

Restaurant Analysis

15. Which cuisine receives the highest number of orders?
16. Which cuisine generates the highest revenue?
17. Which restaurants have the highest ratings?
18. Which restaurants have the longest preparation times?
19. Does preparation time affect delivery time?
20. Which cuisine categories have higher delivery times?

Driver Analysis

21. Does driver experience affect delivery time?
22. Which experience group has the best delivery performance?
23. Does driver rating relate to delivery performance?
24. Which drivers handle the highest number of orders?
25. Which driver groups have higher average delivery times?

Traffic Analysis

26. How does traffic density affect delivery time?
27. Which traffic condition has the highest delivery time?
28. What percentage of orders experience high traffic?
29. Does high traffic increase delivery delays?
30. What is the average ETA under different traffic conditions?

Weather Analysis

31. Which weather condition is associated with longer delivery times?
32. Does bad weather increase delivery delays?
33. Which weather condition has the highest average ETA?

34. What percentage of deliveries occur under difficult weather conditions?

Financial Analysis

35. What is the total order value?
36. What is the average order value?
37. What is the total discount provided?
38. What is the total tax collected?
39. What is the total delivery fee?
40. Which loyalty tier has the highest order value?
41. Which cuisine generates the highest revenue?
42. Does discount level affect order value?

ETA & Delay Analysis

43. What is the average ETA?
44. Which factors have the strongest relationship with delivery time?
45. Which traffic conditions create the most delays?
46. Which weather conditions create the most delays?
47. Does restaurant preparation time affect ETA?
48. Does driver experience affect ETA?
49. Which orders have unusually high delivery times?
50. What factors should be considered for future ETA prediction?

Project Scope

The project includes:

- Data loading
- Data cleaning
- Data validation
- Exploratory data analysis
- Customer analysis
- Restaurant analysis
- Driver analysis
- Traffic analysis
- Weather analysis
- Financial analysis

- ETA analysis
- Outlier detection
- SQL analysis
- Power BI dashboard
- Business insights
- ETA prediction preparation

Technology Stack

Python

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / Google Colab

SQL

- MySQL

Visualization

- Power BI
- Matplotlib
- Seaborn

Data Storage

- CSV
- MySQL

Version Control

- Git
- GitHub

Recommended Project Folder Structure

Food_Delivery_ETA_Analysis/
│

```
├── data/
│   └── food_delivery_orders.csv
│
├── notebooks/
│   └── food_delivery_analysis.ipynb
│
├── sql/
│   └── food_delivery_analysis.sql
│
├── powerbi/
│   └── food_delivery_dashboard.pbix
│
├── images/
│   └── dashboard_screenshots/
│
├── README.md
│
└── requirements.txt
```

Complete End-to-End Project Workflow

Dataset
↓
Load Data
↓
Data Understanding
↓
Missing Value Analysis
↓
Duplicate Analysis
↓
Data Cleaning
↓
Data Validation
↓
Feature Engineering
↓
EDA
↓
Customer Analysis
↓
Restaurant Analysis
↓
Driver Analysis

↓
Traffic & Weather Analysis
↓
Financial Analysis
↓
ETA & Delay Analysis
↓
Outlier Detection
↓
SQL Analysis
↓
Power BI
↓
Business Insights
↓
Recommendations
↓
ETA Prediction Preparation

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load Dataset

```
df = pd.read_csv("food_delivery_orders.csv")

df.head()
```

Check the dataset size:

```
df.shape
```

Expected result:

(50000, number_of_columns)

Step 3: Understand Dataset

df.head()

df.tail()

df.info()

df.columns

df.describe()

For both numerical and categorical variables:

df.describe(include="all")

Step 4: Check Unique Values

df.nunique()

Categorical variable examples:

df["loyalty_tier"].value_counts()

df["cuisine"].value_counts()

df["traffic"].value_counts()

df["weather"].value_counts()

> Replace these column names with the exact names in the dataset.

Step 5: Analyze Missing Values

df.isnull().sum()

Percentage of missing values:

```
missing = df.isnull().sum()

missing_percentage = (missing / len(df)) * 100

pd.DataFrame({
    "Missing Values": missing,
    "Percentage": missing_percentage
})
```

Visualize missing values:

```
plt.figure(figsize=(12,6))

sns.heatmap(df.isnull(), cbar=False)

plt.title("Missing Values")
plt.show()
```

Step 6: Check Duplicate Records

```
df.duplicated().sum()
```

Display duplicates:

```
df[df.duplicated()]
```

Remove duplicates if they are confirmed to be unwanted:

```
df = df.drop_duplicates()
```

Step 7: Standardize Column Names

```
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
    .str.replace(r"[^\w]+", "_", regex=True)
```

)

Check:

df.columns

Step 8: Check Data Types

df.dtypes

Convert numerical columns where required:

```
numeric_columns = [
    "age",
    "rating",
    "preparation_time",
    "subtotal",
    "tax",
    "discount",
    "delivery_fee"
]

for col in numeric_columns:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors="coerce")
```

Step 9: Data Validation

Check negative financial values:

```
financial_columns = [
    "subtotal",
    "tax",
    "discount",
    "delivery_fee"
]

for col in financial_columns:
    if col in df.columns:
        print(col, (df[col] < 0).sum())
```

Check age:

```
if "age" in df.columns:
    print(df["age"].describe())
```

Check ratings:

```
if "rating" in df.columns:
    print(df["rating"].describe())
```

The acceptable rating range should be confirmed from the dataset documentation.

Step 10: Financial Feature Engineering

If the required columns are available, calculate the final order amount.

```
df["calculated_total"] = (
    df["subtotal"]
    + df["tax"]
    + df["delivery_fee"]
    - df["discount"]
)
```

Check the result:

```
df[[
    "subtotal",
    "tax",
    "discount",
    "delivery_fee",
    "calculated_total"
]].head()
```

Step 11: Delivery Feature Engineering

If actual delivery time and ETA are available:

```
df["delay"] = df["actual_delivery_time"] - df["eta"]
```

If delay is measured in minutes:

```
df["delay_minutes"] = (
    df["actual_delivery_time"] - df["eta"]
)
```

Create a delay flag:

```
df["is_delayed"] = np.where(
    df["delay_minutes"] > 0,
    1,
    0
)
```

> Adjust this calculation according to the actual ETA and delivery-time columns.

Step 12: Basic Dataset Analysis

Total orders:

```
total_orders = len(df)

print("Total Orders:", total_orders)
```

Unique customers:

```
if "customer_id" in df.columns:
    print("Unique Customers:", df["customer_id"].nunique())
```

Unique restaurants:

```
if "restaurant_id" in df.columns:
    print("Unique Restaurants:", df["restaurant_id"].nunique())
```

Unique drivers:

```
if "driver_id" in df.columns:
    print("Unique Drivers:", df["driver_id"].nunique())
```

Step 13: Customer Analysis

Customer Loyalty Distribution

```
sns.countplot(data=df, x="loyalty_tier")

plt.title("Orders by Customer Loyalty Tier")
plt.xticks(rotation=45)
plt.show()
```

Customer Age Distribution

```
sns.histplot(df["age"], bins=20, kde=True)

plt.title("Customer Age Distribution")
plt.show()
```

Average Order Value by Loyalty Tier

```
df.groupby("loyalty_tier")["calculated_total"].mean().sort_values(
    ascending=False
)
```

Step 14: Restaurant Analysis

Orders by Cuisine

```
cuisine_orders = (
    df["cuisine"]
    .value_counts()
)

cuisine_orders
```

Visualization:

```
cuisine_orders.plot(kind="bar", figsize=(10,5))

plt.title("Orders by Cuisine")
plt.xlabel("Cuisine")
plt.ylabel("Number of Orders")

plt.show()
```

Step 15: Restaurant Rating Analysis

```
restaurant_rating = (
    df.groupby("restaurant_id")["restaurant_rating"]
    .mean()
    .sort_values(ascending=False)
)
```

Top restaurants:

```
restaurant_rating.head(10)
```

> Replace `restaurant_rating` with the actual rating column.

Step 16: Preparation Time Analysis

Average preparation time:

```
df["preparation_time"].mean()
```

Preparation time by cuisine:

```
df.groupby("cuisine")["preparation_time"].mean().sort_values(
    ascending=False
)
```

Visualization:

```
sns.boxplot(
    data=df,
```

```
    x="cuisine",
    y="preparation_time"
)

plt.xticks(rotation=45)
plt.title("Preparation Time by Cuisine")
plt.show()
```

Step 17: Driver Analysis

Average delivery time by driver experience:

```
df.groupby("driver_experience")["delivery_time"].mean()
```

Average driver rating:

```
df.groupby("driver_experience")["driver_rating"].mean()
```

Driver experience distribution:

```
sns.countplot(
    data=df,
    x="driver_experience"
)

plt.title("Driver Experience Distribution")
plt.xticks(rotation=45)
plt.show()
```

Step 18: Traffic Analysis

Orders by traffic:

```
df["traffic"].value_counts()
```

Average ETA by traffic:

```
df.groupby("traffic")["eta"].mean().sort_values(
    ascending=False
```

```
)
```

Visualization:

```
sns.boxplot(
    data=df,
    x="traffic",
    y="eta"
)

plt.title("ETA by Traffic Condition")
plt.show()
```

Step 19: Weather Analysis

Orders by weather:

```
df["weather"].value_counts()
```

Average ETA by weather:

```
df.groupby("weather")["eta"].mean().sort_values(
    ascending=False
)
```

Visualization:

```
sns.boxplot(
    data=df,
    x="weather",
    y="eta"
)

plt.title("ETA by Weather Condition")
plt.show()
```

Step 20: Financial Analysis

Total revenue/order value:

```
df["calculated_total"].sum()
```

Average order value:

```
df["calculated_total"].mean()
```

Total discount:

```
df["discount"].sum()
```

Total tax:

```
df["tax"].sum()
```

Total delivery fees:

```
df["delivery_fee"].sum()
```

Step 21: Revenue by Cuisine

```
revenue_by_cuisine = (
    df.groupby("cuisine")["calculated_total"]
    .sum()
    .sort_values(ascending=False)
)

revenue_by_cuisine
```

Visualization:

```
revenue_by_cuisine.plot(
    kind="bar",
    figsize=(10,5)
)

plt.title("Revenue by Cuisine")
plt.xlabel("Cuisine")
plt.ylabel("Revenue")

plt.show()
```

Step 22: ETA Analysis

Average ETA:

```
df["eta"].mean()
```

Median ETA:

```
df["eta"].median()
```

Maximum ETA:

```
df["eta"].max()
```

Minimum ETA:

```
df["eta"].min()
```

ETA distribution:

```
sns.histplot(
    df["eta"],
    bins=30,
    kde=True
)

plt.title("ETA Distribution")
plt.xlabel("ETA")
plt.show()
```

Step 23: Delivery Delay Analysis

If delay data is available:

```
df["is_delayed"].value_counts()
```

Percentage of delayed orders:

```
delay_percentage = (
    df["is_delayed"].mean() * 100
)

print("Delayed Orders %:", delay_percentage)
```

Average delay:

```
df.loc[
    df["delay_minutes"] > 0,
    "delay_minutes"
].mean()
```

Step 24: Delay by Traffic

```
df.groupby("traffic")["is_delayed"].mean() * 100
```

This helps identify traffic conditions associated with higher delay rates.

Step 25: Delay by Weather

```
df.groupby("weather")["is_delayed"].mean() * 100
```

This can help determine whether certain weather conditions are associated with increased delivery delays.

Step 26: Delay by Driver Experience

```
df.groupby("driver_experience")["is_delayed"].mean() * 100
```

This can be used to compare delivery performance across driver experience groups.

Step 27: Correlation Analysis

Select available numerical variables:

```
numeric_df = df.select_dtypes(include=np.number)

correlation = numeric_df.corr()
```

Visualize:

```
plt.figure(figsize=(12,8))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()
```

Important relationships to investigate:

- Preparation time vs ETA
- Traffic vs delivery time
- Driver experience vs delivery time
- Order value vs delivery time
- Discount vs order value
- Weather-related variables vs ETA

Correlation should be interpreted as association and not automatically as causation.

Step 28: Statistical Outlier Analysis

IQR Method

```
def find_outliers(df, column):

    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
```

```
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    return df[
        (df[column] < lower_bound) |
        (df[column] > upper_bound)
    ]
```

Example:

```
eta_outliers = find_outliers(df, "eta")

print("ETA Outliers:", len(eta_outliers))
```

Other variables that can be checked:

Order Value
Preparation Time
ETA
Delivery Time
Discount
Delivery Fee
Customer Age
Driver Rating
Restaurant Rating

Step 29: SQL Database Design

The dataset can be imported into MySQL for relational analysis.

Example database:

```
CREATE DATABASE food_delivery_analysis;

USE food_delivery_analysis;
```

Example Table Structure

The actual schema should match the CSV columns.

Customers

```
CREATE TABLE customers (
    customer_id INT PRIMARY KEY,
    age INT,
    loyalty_tier VARCHAR(50)
);
```

Restaurants

```
CREATE TABLE restaurants (
    restaurant_id INT PRIMARY KEY,
    cuisine VARCHAR(100),
    restaurant_rating DECIMAL(3,2),
    preparation_time DECIMAL(10,2)
);
```

Drivers

```
CREATE TABLE drivers (
    driver_id INT PRIMARY KEY,
    driver_experience VARCHAR(50),
    driver_rating DECIMAL(3,2)
);
```

Orders

```
CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    restaurant_id INT,
    driver_id INT,
    weather VARCHAR(50),
    traffic VARCHAR(50),
    subtotal DECIMAL(12,2),
    tax DECIMAL(12,2),
    discount DECIMAL(12,2),
    delivery_fee DECIMAL(12,2),
    eta DECIMAL(10,2)
);
```

> This is an **example relational design**. Match the table definitions to the actual dataset columns before importing the data.

Step 30: SQL Analysis

Total Orders

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Total Customers

```
SELECT COUNT(*) AS total_customers
FROM customers;
```

Orders by Loyalty Tier

```
SELECT
    c.loyalty_tier,
    COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o
    ON c.customer_id = o.customer_id
GROUP BY c.loyalty_tier
ORDER BY total_orders DESC;
```

Top Customers by Order Value

```
SELECT
    customer_id,
    SUM(
        subtotal + tax + delivery_fee - discount
    ) AS total_order_value
FROM orders
GROUP BY customer_id
ORDER BY total_order_value DESC
LIMIT 10;
```

Revenue by Cuisine

```
SELECT
    r.cuisine,
    SUM(
        o.subtotal + o.tax + o.delivery_fee - o.discount
    ) AS revenue
FROM orders o
JOIN restaurants r
    ON o.restaurant_id = r.restaurant_id
GROUP BY r.cuisine
ORDER BY revenue DESC;
```

Average ETA by Traffic

```
SELECT
    traffic,
    AVG(eta) AS average_eta
FROM orders
GROUP BY traffic
ORDER BY average_eta DESC;
```

Average ETA by Weather

```
SELECT
    weather,
    AVG(eta) AS average_eta
FROM orders
GROUP BY weather
ORDER BY average_eta DESC;
```

Driver Performance

```
SELECT
    d.driver_experience,
    AVG(d.driver_rating) AS avg_rating,
    AVG(o.eta) AS avg_eta,
    COUNT(o.order_id) AS total_orders
```

```
FROM drivers d
JOIN orders o
    ON d.driver_id = o.driver_id
GROUP BY d.driver_experience;
```

Restaurant Performance

```
SELECT
    r.restaurant_id,
    r.cuisine,
    r.restaurant_rating,
    AVG(o.eta) AS average_eta,
    COUNT(o.order_id) AS total_orders
FROM restaurants r
JOIN orders o
    ON r.restaurant_id = o.restaurant_id
GROUP BY
    r.restaurant_id,
    r.cuisine,
    r.restaurant_rating
ORDER BY average_eta DESC;
```

Step 31: Power BI Data Preparation

Import the cleaned dataset into Power BI.

Recommended process:

```
Python Cleaning
      ↓
Clean CSV
      ↓
Power BI
      ↓
Power Query
      ↓
Data Model
      ↓
DAX Measures
      ↓
Dashboard
```

Step 32: Power BI Data Model

If the dataset is provided as one order-level table, a simple model can be used.

If the data is split into multiple tables, a star schema can be created.

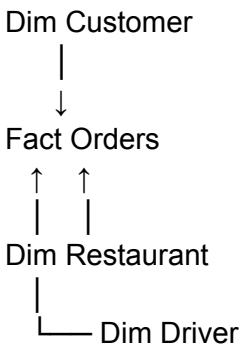
Possible Dimension Tables

Dim Customer
Dim Restaurant
Dim Driver
Dim Date

Possible Fact Table

Fact Orders

Possible relationships:



The exact relationships should be created using the actual primary and foreign keys in the dataset.

Step 33: Power BI DAX Measures

Total Orders

Total Orders =
COUNTROWS(Orders)

Total Customers

```
Total Customers =
DISTINCTCOUNT(Orders[Customer_ID])
```

Total Order Value

```
Total Order Value =
SUMX(
    Orders,
    Orders[Subtotal]
    + Orders[Tax]
    + Orders[Delivery_Fee]
    - Orders[Discount]
)
```

Average Order Value

```
Average Order Value =
DIVIDE(
    [Total Order Value],
    [Total Orders]
)
```

Average ETA

```
Average ETA =
AVERAGE(Orders[ETA])
```

Step 34: Recommended Power BI KPIs

The dashboard can include:

KPI	Description
Total Orders	Number of food delivery orders
Total Customers	Unique customers

Total Restaurants	Unique restaurants
Total Drivers	Unique drivers
Total Order Value	Total customer order value
Average Order Value	Average value per order
Average ETA	Average estimated delivery time
Average Preparation Time	Average restaurant preparation time
Average Driver Rating	Average driver rating
Delayed Orders	Number of delayed orders
Delay Rate	Percentage of delayed orders
Total Discounts	Total discount amount
Total Delivery Fees	Total delivery fee

Step 35: Power BI Dashboard Pages

Page 1 — Executive Overview

KPIs

- Total Orders
- Total Order Value
- Average Order Value
- Average ETA
- Delay Rate

Visuals

- Orders trend
- Revenue by cuisine
- Orders by loyalty tier
- ETA by traffic
- ETA by weather

Page 2 — Customer Analytics

Visuals

- Customer age distribution
- Orders by loyalty tier
- Revenue by loyalty tier
- Average order value by loyalty tier
- Top customers

Slicers

- Loyalty Tier
- Age
- Cuisine

Page 3 — Restaurant Analytics

Visuals

- Orders by cuisine
- Revenue by cuisine
- Restaurant ratings
- Preparation time by cuisine
- Restaurant order volume

Page 4 — Driver Analytics

Visuals

- Driver experience distribution
- Average driver rating
- Average ETA by experience
- Orders by driver
- Driver performance comparison

Page 5 — Logistics & ETA

Visuals

- ETA distribution
- ETA by traffic
- ETA by weather
- Delay rate by traffic
- Delay rate by weather
- Preparation time vs ETA

Page 6 — Financial Analysis

Visuals

- Total order value
- Total discounts
- Total taxes
- Total delivery fees
- Revenue by cuisine
- Average order value by loyalty tier

Page 7 — Delay & Risk Analysis

Visuals

- Delayed vs non-delayed orders
- Delay rate by traffic
- Delay rate by weather
- Delay rate by driver experience
- High-ETA orders
- ETA outliers

Step 36: Possible Business Insights

The actual insights should be generated only after running the analysis.

Possible findings may include:

Customer

- Certain loyalty tiers may generate a larger share of orders.
- Some customer age groups may order more frequently.
- Higher-value orders may be concentrated in specific customer segments.

Restaurant

- Some cuisines may generate higher order volume.
- Certain restaurants may have longer preparation times.
- Preparation time may show a relationship with ETA.

Driver

- Driver experience groups may have different average delivery times.
- Driver ratings can be compared with delivery performance.

Traffic

- Higher traffic conditions may be associated with longer ETAs.
- Delivery delays may increase under heavy traffic.

Weather

- Some weather conditions may be associated with longer delivery times.
- Weather-related operational conditions can be monitored for planning.

Finance

- Certain cuisines or customer groups may contribute more order value.
- Discounts can be analyzed against order value.
- Delivery fees contribute to the final customer payment.

These are **possible analytical findings**, not confirmed results. The final report should use actual values calculated from the dataset.

Step 37: Business Recommendations

Based on the actual analysis, potential recommendations include:

1. Improve ETA Estimation

Use historical:

- Traffic
- Weather
- Preparation time
- Driver experience
- Restaurant performance

to improve ETA calculations.

2. Monitor Restaurant Preparation

Restaurants with consistently high preparation times can be identified and monitored.

3. Improve Driver Allocation

Driver availability and experience can be considered when assigning delivery orders.

4. Monitor High-Traffic Areas

High-traffic conditions can be incorporated into delivery planning.

5. Prepare for Weather Conditions

Historical weather-related delivery patterns can support operational planning.

6. Customer Segmentation

Customer loyalty and order behavior can be used to understand customer segments.

7. Monitor Discounts

Analyze whether discounts increase order value or simply reduce revenue per order.

Step 38: ETA Prediction Preparation

The dataset can also be prepared for a future machine learning model.

Possible Target

ETA

or

Delivery Time

depending on the actual dataset.

Possible Features

Customer Age
Loyalty Tier
Cuisine
Restaurant Rating
Preparation Time
Driver Experience
Driver Rating
Weather
Traffic
Subtotal
Discount
Delivery Fee

Machine Learning Workflow

Raw Data
↓
Data Cleaning
↓
Feature Engineering
↓
Categorical Encoding
↓
Train-Test Split
↓
Model Training
↓
Prediction
↓
Evaluation

Possible regression algorithms:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting
- XGBoost

Model selection should be based on the actual prediction objective and evaluation results.

Step 39: Model Evaluation Metrics

For ETA prediction, common regression metrics include:

MAE

Mean Absolute Error

Measures the average absolute difference between predicted and actual delivery time.

RMSE

Root Mean Squared Error

Penalizes larger prediction errors more heavily.

R^2 Score

Measures the proportion of variation explained by the model.

Step 40: Final Project Deliverables

The project should contain:

✓ Clean Dataset
✓ Python Notebook
✓ EDA
✓ Data Validation
✓ Feature Engineering
✓ SQL Database
✓ SQL Queries
✓ Power BI Dashboard
✓ KPI Analysis
✓ ETA Analysis
✓ Outlier Analysis
✓ Business Insights
✓ Business Recommendations
✓ README

Step 41: Final Report Structure

The final report can be organized as:

1. Introduction
2. Business Problem
3. Project Objectives
4. Dataset Description
5. Data Cleaning
6. Data Validation
7. Exploratory Data Analysis
8. Customer Analysis
9. Restaurant Analysis
10. Driver Analysis
11. Traffic Analysis
12. Weather Analysis
13. Financial Analysis
14. ETA Analysis
15. Outlier Analysis
16. SQL Analysis
17. Power BI Dashboard
18. Key Insights
19. Recommendations
20. Limitations
21. Future Scope
22. Conclusion

Step 42: Project Limitations

Possible limitations include:

- The dataset contains historical observations and may not represent every real-world delivery situation.
- Some operational factors may not be available.
- Actual geographic distance may not be available.
- Real-time traffic information may not be available.
- Real-time driver availability may not be available.
- Actual delivery outcomes should be verified against the available target variable.
- Correlation does not establish causation.
- ETA prediction performance depends on the quality and completeness of the target variable.

Step 43: Future Scope

The project can be extended by:

- Building a machine learning ETA prediction model.
- Predicting delivery delays.
- Adding geographic distance.
- Adding GPS-based driver movement data.
- Adding real-time traffic information.
- Adding real-time weather information.
- Building customer segmentation models.
- Developing restaurant performance scoring.
- Developing driver performance analytics.
- Creating automated Power BI refresh pipelines.

Step 44: GitHub README

Project Title

End-to-End Food Delivery Orders & ETA Logistics Analysis

Project Description

This project analyzes 50,000 food delivery orders to understand customer behavior, restaurant operations, driver performance, environmental conditions, order economics, and delivery performance.

Python is used for data cleaning, exploratory data analysis, feature engineering, and statistical analysis. SQL is used for relational business analysis, while Power BI is used to create an interactive dashboard.

Tools Used

Python
Pandas
NumPy
Matplotlib
Seaborn
MySQL
Power BI
Excel
Git
GitHub

Key Analysis Areas

- Customer behavior
- Loyalty analysis
- Restaurant performance
- Driver performance
- Traffic analysis
- Weather analysis
- Financial analysis
- ETA analysis
- Delay analysis
- Outlier detection

Step 45: Resume Project Description

End-to-End Food Delivery Orders & ETA Logistics Analysis | Python, SQL, Power BI

- Analyzed **50,000 food delivery orders** to study customer behavior, restaurant operations, driver performance, traffic, weather, order value, and delivery performance.
- Cleaned and transformed delivery data using **Python and Pandas**, including missing-value analysis, duplicate detection, data validation, and feature engineering.
- Used **SQL joins, aggregations, and filtering** to analyze customer segments, cuisine performance, driver operations, order value, traffic conditions, and ETA-related metrics.
- Built an interactive **Power BI dashboard** with KPIs and visualizations for order trends, customer loyalty, restaurant performance, driver metrics, financial analysis, and delivery delays.
- Applied statistical analysis and **IQR-based outlier detection** to identify unusual order values, preparation times, ETA values, and delivery-related observations.

Step 46: Short Resume Version

Food Delivery Orders & ETA Logistics Analysis — Python, SQL, Power BI

Analyzed 50K food delivery orders using Python, SQL, and Power BI to evaluate customer behavior, restaurant preparation, driver performance, traffic, weather, financial metrics, and

delivery delays. Built interactive dashboards and performed data cleaning, feature engineering, SQL analysis, and outlier detection to support ETA and logistics analysis.

Step 47: Interview Explanation

Tell me about your project.

> “I worked on an End-to-End Food Delivery Orders and ETA Logistics Analysis project using a dataset of 50,000 food delivery orders. The main objective was to understand customer behavior, restaurant preparation, driver performance, traffic, weather, order value, and delivery performance.
>
> I used Python and Pandas for data cleaning, missing-value analysis, duplicate checking, data validation, exploratory analysis, and feature engineering.
>
> I used SQL to perform business analysis such as customer segmentation, revenue analysis, cuisine performance, driver analysis, and ETA analysis.
>
> Finally, I used Power BI to create an interactive dashboard showing KPIs, order trends, customer loyalty, restaurant performance, driver performance, financial metrics, traffic, weather, and delivery delays.
>
> I also performed statistical outlier analysis to identify unusual delivery and order-related observations. The project can further be extended into a machine learning model for ETA prediction.”

Step 48: Interview Questions

Basic Questions

1. What is the objective of your project?
2. Why did you choose this dataset?
3. How many records are present?
4. What tools did you use?
5. What is ETA?
6. What factors can affect food delivery time?
7. What is exploratory data analysis?
8. How did you handle missing values?
9. How did you identify duplicates?
10. How did you validate the data?

Python Questions

11. Why did you use Pandas?
12. How did you calculate total order value?
13. How did you group data by cuisine?
14. How did you calculate average ETA?
15. How did you identify outliers?
16. What is IQR?
17. Why did you use `groupby()`?
18. What is the difference between `loc` and `iloc`?
19. How did you handle incorrect data types?
20. How did you calculate delay?

SQL Questions

21. What joins did you use?
22. Why did you use GROUP BY?
23. How did you find top customers?
24. How did you calculate revenue by cuisine?
25. How did you calculate average ETA by traffic?
26. What is the difference between WHERE and HAVING?
27. What is a primary key?
28. What is a foreign key?
29. How would you find duplicate orders?
30. How would you find the highest-value order?

Power BI Questions

31. What KPIs did you create?
32. Why did you use Power BI?
33. What is DAX?
34. What is a measure?
35. What slicers did you use?
36. How did you design the dashboard?
37. What is a star schema?
38. How did you create relationships?
39. What is Power Query?
40. How did you improve dashboard readability?

Business Questions

41. How can traffic affect ETA?
42. How can restaurant preparation affect delivery?
43. How can driver experience affect delivery?
44. How can weather affect delivery?

45. How would you reduce delivery delays?
46. How would you improve ETA prediction?
47. How would you identify poor-performing restaurants?
48. How would you analyze customer loyalty?
49. What business KPI is most important for food delivery operations?
50. What would you do if the company gave you real-time delivery data?

Step 49: Final Project Checklist

Data

☐ Dataset loaded
☐ Dataset size checked
☐ Columns checked
☐ Data types checked
☐ Missing values checked
☐ Duplicate records checked
☐ Unique values checked

Cleaning

☐ Column names standardized
☐ Numeric values validated
☐ Missing values handled
☐ Duplicate records handled
☐ Invalid values checked

Analysis

☐ Customer analysis
☐ Restaurant analysis
☐ Driver analysis
☐ Traffic analysis
☐ Weather analysis
☐ Financial analysis
☐ ETA analysis
☐ Delay analysis
☐ Outlier analysis

SQL

☐ Database created
☐ Tables created

☐ Data imported
☐ Joins performed
☐ Aggregations performed
☐ Business queries completed

Power BI

☐ Data imported
☐ Relationships created
☐ DAX measures created
☐ KPI cards created
☐ Charts created
☐ Slicers added
☐ Dashboard completed

Final

☐ Insights documented
☐ Recommendations documented
☐ Limitations documented
☐ README created
☐ GitHub repository prepared
☐ Resume description prepared
☐ Interview explanation prepared

Step 50: Conclusion

The **End-to-End Food Delivery Orders & ETA Logistics Analysis** project provides a complete framework for analyzing a large food delivery dataset containing **50,000 orders**.

The project covers customer behavior, restaurant preparation, driver operations, traffic, weather, financial metrics, ETA, and delivery delays.

Using **Python**, the data can be cleaned, validated, explored, and transformed. Using **SQL**, business questions can be answered through filtering, joins, aggregations, and relational analysis. Using **Power BI**, the results can be presented through interactive dashboards and KPIs.

The project also provides a strong foundation for future **ETA and delivery-delay prediction using machine learning**.

The final objective is to convert raw food delivery data into useful operational information that can help understand delivery performance, customer behavior, restaurant operations, and logistics efficiency.